



UNIVERSITY OF
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Introduction to Blockchain 2026

Project Work - Final Report

Hotel Booking Distributed Application

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Table of Contents

Abstract.....	3
1. Introduction and Conceptual Plan.....	4
1.1 Overview	4
1.2 Motivation	4
1.3 Blockchain Concepts and Implementation	4
1.4 Challenges	4
2. Technical Implementation of the Project	4
2.1 Introduction	4
2.2 Logical Design and System Architecture	4
2.3 Tools and Software	5
3. Blockchain Concepts Demonstrated in the Project.....	6
3.1 Smart Contracts	6
3.2 Transactions	6
3.3 Decentralized Application (DApp)	6
4. Project features and detailed elaboration of the blockchain concept	6
4.1 Project features	6
4.2 Detailed elaboration of the blockchain concept	8
5. Real world Use case for the Hotel Booking application.....	9
5.1. Immutable Reservation Records	9
5.2. Smart Contract Automated Booking Management	9
5.3. Transparent Pricing System	9
6. Codebase of the project	9
7. Deployed Application.....	9
8. The final presentation.....	9
9. Video Demonstration of the Application	10

Abstract

Blockchain technology is a decentralized ledger system that enables secure, transparent, and immutable recording of transactions. This thesis explores blockchain fundamentals, smart contracts, decentralized applications (DApps), system architecture, and implementation using Ethereum-based tools. The project demonstrates how blockchain can be applied to build trustless systems without centralized intermediaries.

1. Introduction and Conceptual Plan

1.1 Overview

Blockchain is a distributed ledger technology that stores data across multiple nodes without relying on a central authority. Each block contains transaction data, a timestamp, and a cryptographic link to the previous block.

1.2 Motivation

The motivation for blockchain arises from the need for decentralized trust, secure data storage, transparency, and elimination of intermediaries in digital systems.

1.3 Blockchain Concepts and Implementation

Blockchain consists of blocks, cryptographic hashing, consensus mechanisms, and smart contracts. Ethereum enables programmable blockchain functionality through the Ethereum Virtual Machine (EVM).

1.4 Challenges

Blockchain faces challenges including scalability limitations, high energy consumption, regulatory uncertainty, storage inefficiency, and smart contract vulnerabilities.

2. Technical Implementation of the Project

2.1 Introduction

The project demonstrates a decentralized application that simulates blockchain transactions and smart contract execution.

2.2 Logical Design and System Architecture

The system consists of three layers: frontend DApp interface, smart contract layer, and blockchain network layer. Users interact through a web interface connected to Ethereum.

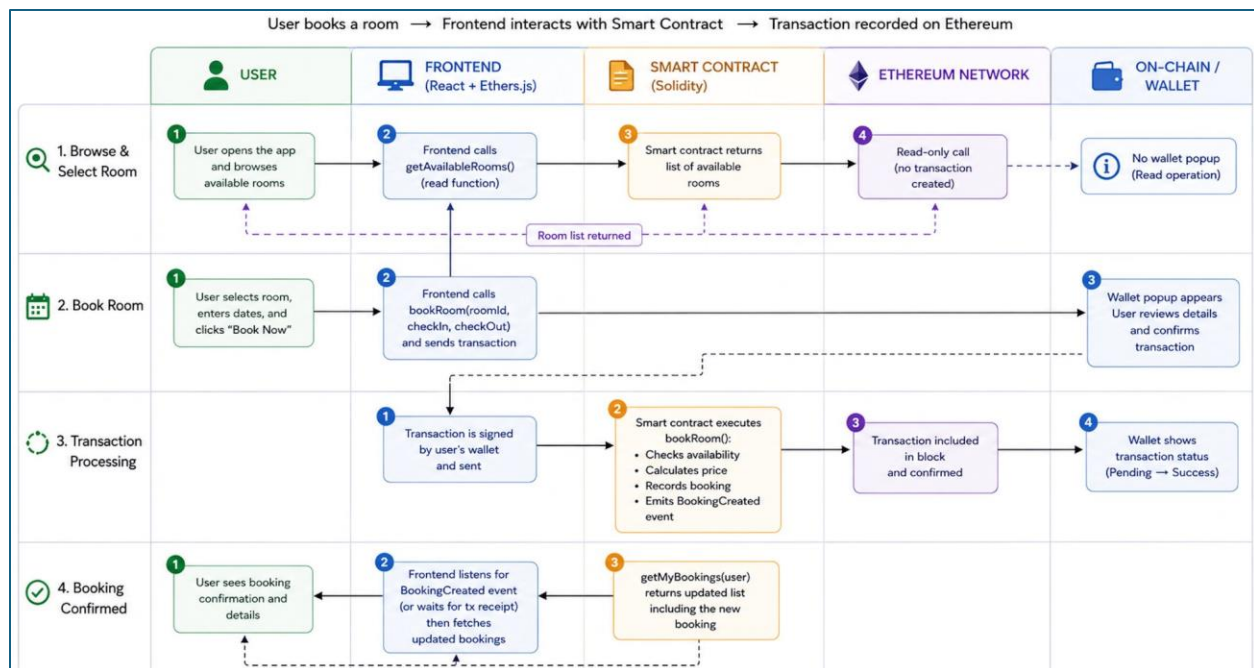


Figure 1: Logical design of the application

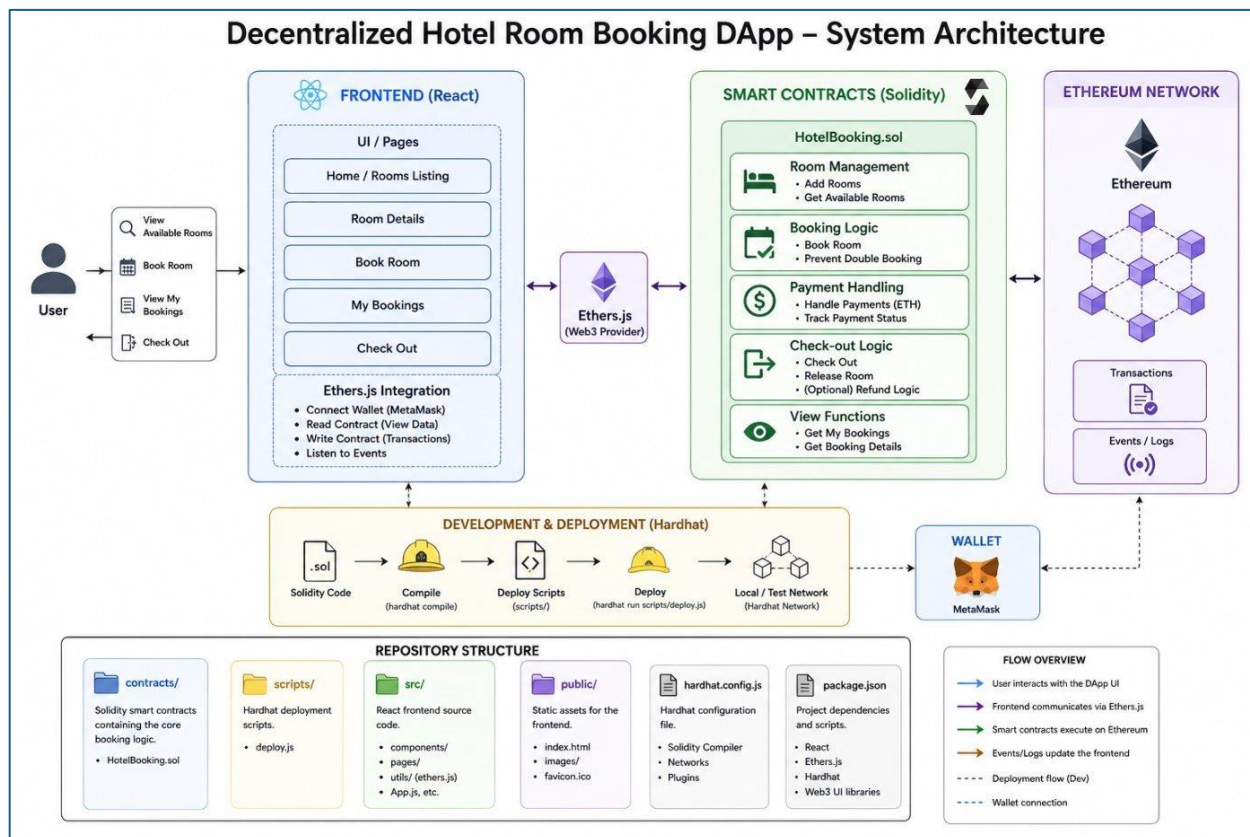


Figure 2: System Architecture

2.3 Tools and Software

Tools used include Ethereum, Solidity, VS Code IDE, MetaMask, Node.js, and Web3.js.

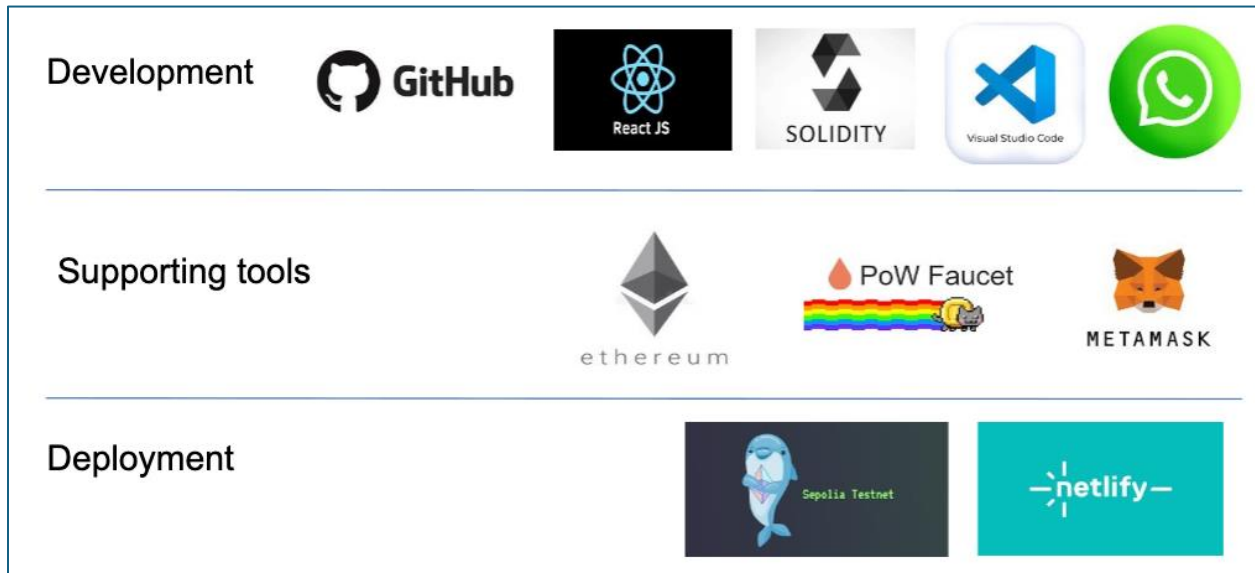


Figure 3: Technical stack

3. Blockchain Concepts Demonstrated in the Project

3.1 Smart Contracts

Smart contracts are self-executing programs deployed on blockchain that automate transaction logic.

3.2 Transactions

Transactions are cryptographically signed data structures that are validated and recorded on blockchain.

3.3 Decentralized Application (DApp)

A DApp is an application that runs on blockchain infrastructure instead of centralized servers.

4. Project features and detailed elaboration of the blockchain concept

4.1 Project features

The project is a decentralized hotel booking application (DApp) built on Ethereum that allows users to browse rooms, book accommodations, and confirm reservations through blockchain transactions and smart contracts.

Main features include:

1. **Room Browsing and Availability Checking**
 - a. Users can browse hotel rooms and check room availability in real time.

- b. Read-only blockchain queries are used to retrieve room data without creating transactions.
- 2. **Blockchain-Based Booking**
 - a. Users can book rooms directly through smart contracts.
 - b. Booking data is recorded immutably on the blockchain.
- 3. **Wallet Integration**
 - a. Users connect crypto wallets to authorize and sign transactions securely.
 - b. Digital signatures verify transaction authenticity.
- 4. **Smart Contract Automation**
 - a. Smart contracts automatically:
 - i. validate room availability,
 - ii. calculate booking prices,
 - iii. store reservation records,
 - iv. emit booking confirmation events.
- 5. **Transparent Transaction Processing**
 - a. Every booking transaction is visible and verifiable on the blockchain network.
 - b. Users can track transaction states such as pending and confirmed.
- 6. **Event-Driven Frontend Updates**
 - a. The frontend listens to blockchain events and updates booking information automatically after transaction confirmation.
- 7. **Decentralized Architecture**
 - a. The application removes the need for centralized intermediaries and relies on blockchain consensus and distributed ledger technology.

4.2 Detailed elaboration of the blockchain concept

1. **Blockchain:** Blockchain is a distributed digital ledger that stores transactions in linked blocks secured by cryptography. Once data is recorded, it becomes difficult to modify, ensuring immutability and transparency. In this project: booking transactions, reservation records, and smart contract interactions are permanently stored on the blockchain.

2. **Smart Contracts:** Smart contracts are self-executing programs running on the blockchain. In the hotel booking system, the smart contract manages room reservations, records reservations, and emits confirmation of events automatically. The business logic is executed without requiring a centralized booking authority.

3. **Ethereum Virtual Machine (EVM):** The Ethereum Virtual Machine (EVM) executes smart contract code across all Ethereum nodes. Every booking operation is processed by the EVM, ensuring deterministic and consistent execution throughout the network.

4. **Gas Mechanism:** Gas is the computational fee required to execute blockchain operations. In this project, browsing room data uses read-only calls and consumes no transaction of gas, and booking a room requires gas fees because the blockchain state changes. Gas prevents spam and compensates validators for processing transactions.

5. Digital Signatures

Blockchain transactions are authenticated using public-key cryptography. When a user books a room, the wallet signs the transaction using the user's private key, and the blockchain verifies the signature using the corresponding public key. This guarantees authenticity and ownership verification.

6. **Consensus Mechanism:** The blockchain network uses a consensus algorithm to ensure all nodes agree on the same ledger state. For Ethereum, the current mechanism is Proof of Stake (PoS), where validators confirm transactions and create blocks. Consensus guarantees that booking records cannot be fraudulently altered.

7. **Immutability:** Once a booking transaction is confirmed and added to a blockchain, the reservation data cannot easily be changed or deleted. If corrections are required, new transactions must be added instead of modifying existing records.

8. **Transparency and Trustlessness:** All confirmed booking transactions are publicly verifiable on the blockchain, users do not need to trust a centralized booking provider because the smart contract logic is transparent, transaction validation is decentralized, and blockchain consensus ensures integrity.

9. **DApp Architecture:** The project follows a decentralized application architecture consisting of:

- Frontend (React + Ethers.js)
- Smart Contracts (Solidity)
- Ethereum Network
- Crypto Wallet Integration

The frontend communicates with smart contracts through blockchain APIs and wallet providers.

5. Real world Use case for the Hotel Booking application

5.1. Immutable Reservation Records

Traditional Problem: In centralized systems, booking data may be altered, records may be lost, or disputes may occur regarding reservations and payments.

Blockchain-Based Solution: Every booking transaction is permanently stored on the blockchain. And once the transaction has been confirmed, the reservation records become immutable, booking history becomes auditable, and tampering becomes extremely difficult.

Benefits: Prevents reservation fraud, reduces booking disputes, creates trusted audit trails and improves accountability

5.2. Smart Contract Automated Booking Management

Traditional Problem: Hotel management systems often require manual verification, staff approval, payment reconciliation, and centralized database operations.

Blockchain-Based Solution: Smart contracts automate room availability validation, reservation confirmation, payment handling

Benefits: The app can reduce human errors, 24/7 automated operation, faster processing and so, can lower administrative costs

5.3. Transparent Pricing System

Traditional Problem: Customers sometimes face hidden fees, unexpected taxes, dynamic price manipulation, or inconsistent pricing across platforms.

Blockchain-Based Solution: Smart contracts publicly define booking rules, pricing logic

Benefits: By clearly knowing the price beforehand, the app can increase customer trust, reduced pricing disputes, and transparent fee structure.

6. Codebase of the project

GitHub Repository (main branch): <https://github.com/anninh-nguyen/Hotel-Dapp>

7. Deployed Application

Web hosting Netlify: <https://hotel-dap.netlify.app/>

8. The final presentation

The report on GitHub: https://github.com/anninh-nguyen/Hotel-Dapp/blob/main/docs/IBC2026-Final_Presentation.pdf

9. Video Demonstration of the Application

YouTube Video: <https://www.youtube.com/watch?v=GF9XqH-DaIl>

